National University of Computer and Emerging Sciences

Islamabad Campus

Generative AI (AI4009)

Course Instructor(s):

Dr. Akhtar Jamil

Sections: A B C

Sessional-I Exam

Total Time (Hrs): 1

Total Marks: 50

Total Questions: 3

Date: Feb 25, 2026

Roll No ______ **Course Section** ______ **Student Signature** ______

Do not write below this line.

Attempt all the questions.

[CLO 1: Explain and interpret the foundational principles of generative AI, including, neural architectures (CNNs, RNNs, Transformers), and their role in generative modeling.]

Question No. 1. MCQ [1 x 20 = 20]

Answer the MCQs on the given answer sheet attached at the end of the question paper. Answers marked on the question paper will not be evaluated.

1. If the discriminator becomes too strong early in training, what is the most likely consequence?
A. Generator gradients become unstable
B. Generator produces more diverse outputs
C. Generator training accelerates
D. Loss function becomes convex

2. A neural network uses a LeakyReLU activation with slope 0.5. For the input vector: $x = [4, -2, 3]$, What will be the output after applying the activation function?
A) [4, 0, 3]
B) [4, −2, 3]
C) [2, −1, 1.5]
D) [4, −1, 3]

3. Why is CycleGAN considered a solution when paired training data is not available in image-to-image translation tasks?
A) It does not require any training samples to learn mappings between domains.
B) It learns mappings between two domains using unpaired datasets by enforcing cycle-consistency constraints.
C) It operates only on grayscale images to simplify training.
D) It uses a single generator shared across both domains.

4. Why do Variational Autoencoders (VAEs) often generate images that appear blurrier than those in the original dataset?
A. Adversarial loss smooths out fine details
B. Latent space regularization removes sharp edges
C. Encoders output nearly identical latent codes
D. Reconstruction loss averages variations, reducing sharpness

5. How does Mini-Batch Discrimination solve the problem of Mode Collapse?
A) By making the Discriminator faster so it can see more images per second.
B) By allowing the Discriminator to compare the diversity of a batch, penalizing the Generator if the generated samples are too similar to each other.
C) By using L1 and L2 distances to ensure the Generator produces images that are pixel-perfect copies of real data.
D) By ensuring the Generator only produces one image at a time to prevent confusion.

6. If the discriminator becomes perfect early in GAN training and correctly classifies all generated samples as fake, what problem does this create for training the generator?
A) The generator loss becomes zero and stops updating.
B) The discriminator provides little useful gradient signal for the generator to learn from.
C) The generator automatically switches to another training method.
D) The optimizer stops updating the generator weights.

7. Why does the CycleGAN decoder use Transposed Convolution layers?
A) To normalize the data across each channel independently.
B) To increase the spatial dimensions back to the original size after the encoder reduced them.
C) To calculate the adversarial loss for domain .
D) To apply the Tanh activation function.

8. Why do Mini-Batch GANs reduce mode collapse compared to standard GAN training?
A. They force the generator to memorize training data.
B. **They allow the discriminator to compare relationships among multiple samples.**
C. They eliminate all gradient noise during optimization.
D. They increase the generator's parameter space.

9. The discriminator loss in GANs is defined as:

$$L_D = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(D(x_i)) + (1-y_i)\log(1-D(G(z_i)))]$$

For a mini-batch of $N = 2$ with just two classes:

- Sample 1: real data x_1, $y_1 = 1$, $D(x_1) = 0.8$
- Sample 2: generated data $G(z_2)$, $y_2 = 0$, $D(G(z_2)) = 0.3$

What is the discriminator loss L_D(use natural log, round to 3 decimal places)?
A. 0.123
B. 0.290
C. 0.456
D. 0.678

10. In GANs, we should not use a pretrained Discriminator to improve the quality of the Generator and reduce training time.
A. True
B. False

11. Which one of the following can solve mode collapse problem?

A. Mini-batch discrimination
B. Multi-label Supervised training
C. **All of the above**
D. None of the above

12. What is the purpose of pooling in neural networks?

A. To increase the depth of the feature maps.
B. To reduce the spatial dimensions of the input.
C. To perform element-wise multiplication between feature maps.
D. To increase the spatial dimensions of the input.

13. For which of the following problems, RNN is more suitable:

A. Time series forecasting
B. Sentiment analysis on text data
C. Image classification
D. Both A and B

14. What cost is introduced when using a Maxout layer instead of a ReLU layer in a neural network?

A) It increases the number of learnable parameters.
B) It prevents the model from learning non-linear patterns.
C) It restricts the network to only categorical inputs such as one-hot vectors.
D) It requires tanh activation in later layers.

15. You are given a very large dataset where most of the data is unlabeled and only a small portion is labeled. `Which of the following approaches is most suitable for this scenario?

A. Train a standard feedforward ANN directly on the small, labeled portion
B. Use a recurrent neural network trained only on the labeled portion
C. **First train an autoencoder on the unlabeled data, then use the encoder to train a classifier on the small, labeled set**
D. Train a GAN, since GANs do not require large amounts of data

16. Why does alternating training between the Generator and Discriminator make convergence in GAN training difficult?

A) Because both networks update their parameters based on feedback that keeps changing as the other network improves.
B) Because alternating training prevents backpropagation from being applied correctly.
C) Because only one network can be trained per epoch in deep learning frameworks.
D) Because alternating updates force the Generator to use different loss functions each step.

17. If the learning rate is set too small during training, what is the most likely effect on the learning process?

A. The model will converge very quickly.
B. The model will learn very slowly and take many iterations to converge.
C. The model will suffer from exploding gradients.
D. None of the above.

18. Consider a model that learns the following joint probability distribution for three sequential inputs X_1, X_2, X_3 and the label Y:

$$P(X_1, X_2, X_3, Y) = P(Y) \cdot P(X_1|Y) \cdot P(X_2|X_1, Y) \cdot P(X_3|X_2, X_1, Y)$$

Is this model generative or discriminative?

A. **Generative**
B. Discriminative
C. None

19. Given the following probabilities, what is the joint probability $P(X = \text{Generative}, Y = \text{AI})$?
$P(X = \text{Discriminative}) = 0.4$, $P(X = \text{Generative}) = 0.6$,
$P(Y = \text{AI} \mid X = \text{Discriminative}) = 0.3$, $P(Y = \text{AI} \mid X = \text{Generative}) = 0.5$

A. 0.20
B. 0.24
C. **0.30**
D. 0.50

20. In the context of training neural networks, what is the role of the momentum term in gradient descent?

A. It increases the learning rate dynamically when gradients are small
B. It helps escape local minima by adding a fraction of the previous update to the current update
C. It reduces overfitting by penalizing large weights
D. It normalizes gradients to keep them within a fixed range

[CLO 1: Explain and interpret the foundational principles of generative AI, including Bayesian learning, neural architectures (CNNs, RNNs, Transformers), and their role in generative modeling.]

Question No 2. Write short answers to the following questions. [3 x 5=15]

1) In a Variational Autoencoder (VAE), the latent vector is sampled using the reparameterization trick by using μ, σ, and ε. Given μ = [2, 0, −2], σ = [2, 4, 6], and ε = [0.5, 0.5, 0.5], sample a latent vector z.

   Answer:
   z = [3, 2, 1]

2) What happens to the output of a GAN if noise prior is kept constant but the condition is changed?. If the model is trained properly, the generated output will change according to the condition y (such as class label or attributes).
   Since z is fixed, general visual characteristics like pose, layout, thickness, or style often remain similar, while the class-related content changes.

3) How does compressing a 3072-dimensional input into 128 dimensions filter out unnecessary details?

   Compressing a 3072-dimensional input into 128 dimensions forces the model to keep only the most important patterns needed to reconstruct the image.
   Since the smaller representation cannot store every pixel detail, it learns to preserve general

structure and meaningful features while discarding noise, small variations, and redundant information.
This is why the compressed representation filters out unnecessary details.

4) Why is Instance Normalization preferred in CycleGAN over Batch Normalization?
In image translation, each image has its own unique features; normalizing per-instance is more robust than per-batch

5) In a Variational Autoencoder (VAE), what happens if the KL divergence weight (β) is set too high or too low?

In a Variational Autoencoder (VAE), adjusting the KL divergence weight is a balancing act between accurately reconstructing data and maintaining a structured latent space. The consequences of setting this weight incorrectly are as follows:
• Weight set too low: When the weight is too low, the model places too much emphasis on the reconstruction loss and may ignore the latent space structure. This leads to a situation where there is no information in z (the latent code is ignored), resulting in blurry average images during reconstruction.
• Weight set too high: If the weight is too high, the model prioritizes the KL divergence term over reconstruction, which can lead to the model ignoring the details of the data. In this scenario, the latent code is not compressed properly, leading to too much information in z and resulting in when attempting to sample from the latent space. garbage images

[CLO 1: Explain and interpret the foundational principles of generative AI, including Bayesian learning, neural architectures (CNNs, RNNs, Transformers), and their role in generative modeling.]

Question No 3: **[3x5= 15 marks]**

1) You are using semantic hashing for image de-duplication as described in the DCGAN paper. Three encoded images obtained after passing through an autoencoder are shown below. Using Euclidean distance with a threshold T = 5, determine which two images should be kept, and which image should be removed as a near-duplicate. Show all steps for your calculations.

$$I_1 = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \quad I_2 = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix} \quad I_3 = \begin{bmatrix} 8 & 7 \\ 6 & 9 \end{bmatrix}$$

$$I_1 = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix} \quad I_2 = \begin{bmatrix} 2 & 3 \\ 4 & 6 \end{bmatrix} \quad I_3 = \begin{bmatrix} 8 & 7 \\ 6 & 9 \end{bmatrix}$$

$$d(I_1, I_2) = \sqrt{(2-2)^2 + (3-3)^2 + (4-4)^2 + (5-6)^2} = \sqrt{1} = 1$$

$$d(I_1, I_3) = \sqrt{(-6)^2 + (-4)^2 + (-2)^2 + (-4)^2} = \sqrt{72} \approx 8.49$$

$$d(I_2, I_3) = \sqrt{(-6)^2 + (-4)^2 + (-2)^2 + (-3)^2} = \sqrt{65} \approx 8.06$$

Since $T = 5$:

$$\text{Keep } I_1, I_3 \qquad \text{Remove } I_2$$

2) Suppose the input z has dimension 10. We want to project it through a fully connected (dense) layer so that it can be reshaped into a 2 × 2 × 3 feature map. How many parameters does this fully connected layer need to have (without any biases)?

For 2 × 2 × 3 = 12 outputs

- Input dimension = 10
- Output dimension = 12 (for 2 × 2 × 3)
- Parameters in weight matrix = 10 × 12 = 120
- **Total = 120 parameters**

3) Consider two discrete probability distributions $P$and Qdefined over the same events:

$$P = [0.5,\ 0.3,\ 0.2], Q = [0.4,\ 0.4,\ 0.2]$$

Compute the Kullback–Leibler (KL) Divergence**:**

$$D_{KL}(P \parallel Q) = \sum_i P(i)\ \log \frac{P(i)}{Q(i)}$$

(use natural logarithm, base e).

$$D_{KL}(P \parallel Q) = 0.5\log \frac{0.5}{0.4} + 0.3\log \frac{0.3}{0.4} + 0.2\log \frac{0.2}{0.2}$$
$$= 0.5\log(1.25) + 0.3\log(0.75) + 0.2\log(1)$$
$$= 0.5(0.2231) + 0.3(-0.2877) + 0$$
$$= 0.11155 - 0.08631 = 0.0252$$

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Answer Sheet MCQs

Fill the correct option. Only one option must be selected. Selection of multiple options or overwriting will result in ZERO marks.

Name: ________________________________ **Roll No.** ________________